

Adding in binary

Numbers written in binary form can be added together in a similar way to the decimal system, and column addition may be done like this:

$$\begin{array}{r}
 \text{Fours} \quad \text{Twos} \quad \text{Ones} \\
 111 \\
 + 101 \\
 \hline
 \end{array}$$

(4 + 2 + 1 in decimal)
(4 + 0 + 1 in decimal)

$$\begin{array}{r}
 1 + 1 = 2 \\
 \text{(this is 10 in binary)} \\
 111 \\
 + 101 \\
 \hline
 0
 \end{array}$$

1 ← carry 1

$$\begin{array}{r}
 1 + 0 + 1 = 2 \\
 \text{(this is 10 in binary)} \\
 111 \\
 + 101 \\
 \hline
 00
 \end{array}$$

1 ← carry 1

$$\begin{array}{r}
 \text{the answer is 12} \\
 \text{in the decimal system} \\
 \text{(8 + 4 + 0 + 0 = 12)} \\
 111 \\
 + 101 \\
 \hline
 1100
 \end{array}$$

1 ← carry 1 and place in the eights column

Align the numbers under their correct place-value columns as in the decimal system. It may be helpful to write in the column headings when first learning this system.

Add the ones column. The answer is 2, which is 10 in binary. The twos are shown in the next column so carry a 1 to the next column and leave 0 in the ones column.

Now the digits in the twos column are added together with the 1 carried over from the ones column. The total is 2 again (10 in binary) so a 1 needs to be carried and a 0 left in the twos column.

Finally, add the 1s in the fours column, which gives us 3, (11 in binary). This is the end of the equation so the final 1 is placed in the eights column.

Subtracting in binary

Subtraction works in a similar way to the decimal system but “borrows” in different units to the decimal system—borrowing by twos instead of tens.

$$\begin{array}{r}
 \text{Eights} \quad \text{Fours} \quad \text{Twos} \quad \text{Ones} \\
 1101 \\
 - 11 \\
 \hline
 \end{array}$$

(8 + 4 + 1 in decimal)
(2 + 1 in decimal)

$$\begin{array}{r}
 \text{it is helpful} \\
 \text{to add zeros} \\
 1101 \\
 - 0011 \\
 \hline
 0
 \end{array}$$

subtract the units

$$\begin{array}{r}
 \text{we put a 2 in the twos column} \\
 \text{because the borrowed number} \\
 \text{represents 2 lots of twos} \\
 0 \quad 2 \\
 1101 \\
 - 0011 \\
 \hline
 10
 \end{array}$$

the answer is 10 in the decimal system
(8 + 0 + 2 + 0 = 10)

$$\begin{array}{r}
 0 \quad 2 \\
 1101 \\
 - 0011 \\
 \hline
 1010
 \end{array}$$

The numbers are written in their correct place-value columns as in the decimal system.

Add zeros so that there are the same number of digits in each column. Then begin by subtracting the ones column: 1 minus 1 is 0, so place a 0 in the answer space. Now move on to the twos column on the left.

The lower 1 cannot be subtracted from the 0 above it so borrow from the fours column and replace it with a 0. Then put a 2 above the twos column. Subtract the lower 1 from the upper 2. This leaves 1 as the answer.

Now subtract the digits in the fours column, which gives us 0. Finally, in the eights column we have nothing to subtract from the upper 1, so 1 is written in the answer space.